ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q3

Question 1

(a) f_x = 3200 px. FOV = 160 mm. 1 pixel = 0.125 mm.

(b) Answer: 0.122 N.

(c) $F = e^{AT} = e^{-0.2} = 0.8187$. $G = A^{-1}(F - I)B = (0.8187 - 1)/(-2) = 0.0906$.

Hence the exact model is $x_{k+1} = \mathbf{0.0906\ x_k + 0.8187\ u_k}$.

Question 2

(a) States $x_1 = \theta$, $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$; $\dot{x}_2 = (1/J)[\tau - b\ x_2 - mg\ell \sin x_1]$.

Equilibira ($\tau = \tau_0$, $\dot{x} = 0$): $x_2 = 0$ and $mg\ell \sin \theta_e = \tau_0$. Since $|\sin \theta| \leq 1$, an equilibrum exists iff $|\tau_0| \leq mg\ell = 2.4525$ N·m.

(b) $\delta\dot{x} = A\,\delta x + B\,\delta\tau$, $A = \partial f/\partial x$, $B = \partial f/\partial \tau$ at $\theta_e = 60°$, $x_{2e} = 0$.

$A = [0,\ 1;\ -(mg\ell/J)\cos\theta_e,\ -b/J] = [0,\ 1;\ -122.6\times0.5,\ -2.5] = \mathbf{[0,\ 1;\ -61.31,\ -2.5]}$.

$B = [0;\ 1/J] = \mathbf{[0;\ 50]}$.

(c) At rest $\dot{\theta} = \ddot{\theta} = 0$, so the PD torque must balance gravity: $K_p(\theta_d - \theta_{ss}) = mg\ell \sin \theta_{ss}$.

If $\theta_{ss} = \theta_d = 90°$ the left side is 0 but the right side is $mg\ell \sin 90° = 2.4525 \neq 0$: contradiction. So the joint cannot rest at θ_d; the gravity torqe needs a non-zero error to produce the holding torque.

Solve $50(\pi/2 - \theta_{ss}) = 2.4525 \sin \theta_{ss}$ numerically: $\theta_{ss} \approx 1.5218$ rad = 87.19°. Error ≈ 90° − 87.19° = **2.81°** (joint sags below θ_d).

(d) With compensation, at rest: $K_p(\theta_d - \theta_{ss}) + mg\ell \sin \theta_{ss} = mg\ell \sin \theta_{ss} \Rightarrow \theta_{ss} = \theta_d$: **steady-state error = 0**.

Closed loop: $J\ddot{\theta} + (b + K_d)\dot{\theta} + K_p(\theta - \theta_d) = 0$; gravity cancels. Without compensation the linearised stiffness is $K_p + mg\ell \cos \theta_d$, and at $\theta_d = 90°$ $\cos \theta_d = 0$, so the stiffness is K_p either way. **Poles unchanged** (specific to 90°; at other angles they would shift).

Question 3

(a) (1) Differentiation is a high-pass operation (gain ∝ ω), so encoder measurement/quantisation noise is amplifed, by about 1/T = 1000 at 1 kHz.

(2) Encoder quantisation: Δθ is a whole number of counts, so $\hat{\omega} = \Delta\theta/T$ moves in steps of (count size)/T; at low speed the estimate is a coarse staircase of zeros and spikes.

(b) $y[k] = 0.0591\,(u[k] + u[k-1]) + 0.8818\,y[k-1]$. DC gain = 1.

(c) $L = [18,\ 64]^T$.

(d) Advantage of the obsever: it uses the plant model and the applied input u, so its velocity estimat has (almost) no phase lag; the 20 Hz filter on differenced data always delays by about $1/(2\pi f_c) \approx 8$ ms. It can also reconstruct states that cannot be obtained by differencing (e.g. disturbance, when augmented).

Filter is safer when the model is uncertain or changes (unknown payload, friction, contact, acuator saturation): the observer then gives a biased or poorly-behaved estimate while the model-free fliter still follows the measured data.